Placing the New Tables: Incremental Diagram Layout with a Routed Objective and an Honest Control

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Abstract

Every mainstream database tool lays out its entity–relationship diagrams badly, and the author of this paper spent about an hour re-drawing one 44-table diagram by hand after each reverse-engineering for years. The underlying task is NP-complete, so the tools run heuristics, but the heuristics fail at a specific, avoidable place: they optimize straight center-to-center edges while drawing orthogonal connectors, so the quantity they improve is not the quantity the reader sees. We formulate the practical version of the problem, which is incremental: a migration adds k tables to a diagram whose other tables must not move, because a reader who knows where a table sits should still find it there. The search space is $2k$ variables, the objective reads orthogonally routed connectors, and the router is calibrated against the one certain fact available: a human-maintained layout that achieved zero crossings and zero edges under tables on screen. Four stochastic searches, run at identical budgets against uniform random sampling and judged by an exact sign test on paired seeds, place the tables of **[PENDING: n]** real consecutive migrations of a production schema, each with a human-accepted answer, under a pre-registered analysis. **[PENDING: primary result sentence]**. A finding we did not seek: under the routed objective the searches outscore the human's own layout, which measures the objective's blindness to semantic grouping rather than any superiority of the tool, and we argue this gap, between what layout objectives score and what maintainers optimize, is why automatic diagram layout has stayed bad in practice.

1 Introduction

[PENDING: introduction]

2 The problem

[PENDING: formulation]

3 The objective is the medium

[PENDING: objective]

4 The searches and the control

[PENDING: library]

*Independent researcher. Code, data and derived tables: `https://github.com/Anode1/cjitter`.

5 The benchmark

[PENDING: benchmark]

6 Pre-registered experiment

[PENDING: results]

7 What the human comparison measures

[PENDING: discussion]

8 Related work

[PENDING: related]

9 Conclusion

[PENDING: conclusion]